Econometrics Project Proposal

Fuel Price Differences in France

Group 3

Research Question and Economic Motivation

What factors are associated with differences in fuel prices across gas stations in France?

Fuel prices directly affect transportation costs faced by households and firms. However, prices vary across fuel types, stations and geographical areas. The objective of this project is to investigate which observable factors are associated with these price differences.

Data

We use the French government dataset *Prix des carburants en France – flux quotidien*, available from the French government's open-data platform:

`data.gouv.fr - Prix des carburants en France`

The raw file contains **76,635 rows** and **26 variables**, including fuel prices and types, station identifiers and locations, geographical information, services, opening hours, fuel shortages and station closures.

Among the raw rows, **71,224** contain both a fuel type and a price. Because the raw structure contains repeated rows for some station–fuel combinations, we checked whether these repetitions contained different prices. They did not. We therefore retain one observation per station and fuel type.

Final analysis sample: 29,802 station–fuel observations | 8,999 stations

The dependent variable is the **fuel price**. The main explanatory variables considered are fuel type, geographical location and observable station characteristics.

Preliminary Descriptive Evidence

Figure 1 reports descriptive statistics for prices across the six fuel types in the analysis sample.

```
=== DESCRIPTIVE STATISTICS ===
              N   Mean    Std    Min  Median    Max
Carburant
E85        3770  0.895  0.080  0.794   0.868  2.249
GPLc       1488  1.052  0.086  0.600   1.041  2.191
E10        6577  2.182  0.094  1.880   2.185  2.749
SP95       2743  2.226  0.093  1.169   2.239  2.740
SP98       6745  2.285  0.115  1.255   2.289  2.929
Gazole     8479  2.415  0.091  1.530   2.422  2.900
```

Figure 1: Descriptive statistics of fuel prices by fuel type

The descriptive statistics reveal substantial differences in price levels across fuel types. Mean prices range from approximately €0.895/L for E85 to €2.415/L for Gazole. Dispersion also differs across

fuel categories, suggesting that controlling for fuel type is important when studying price differences across stations.

To examine geographical variation without mechanically comparing different fuel products, Figure 2 focuses on Gazole and compares its average price across regions.

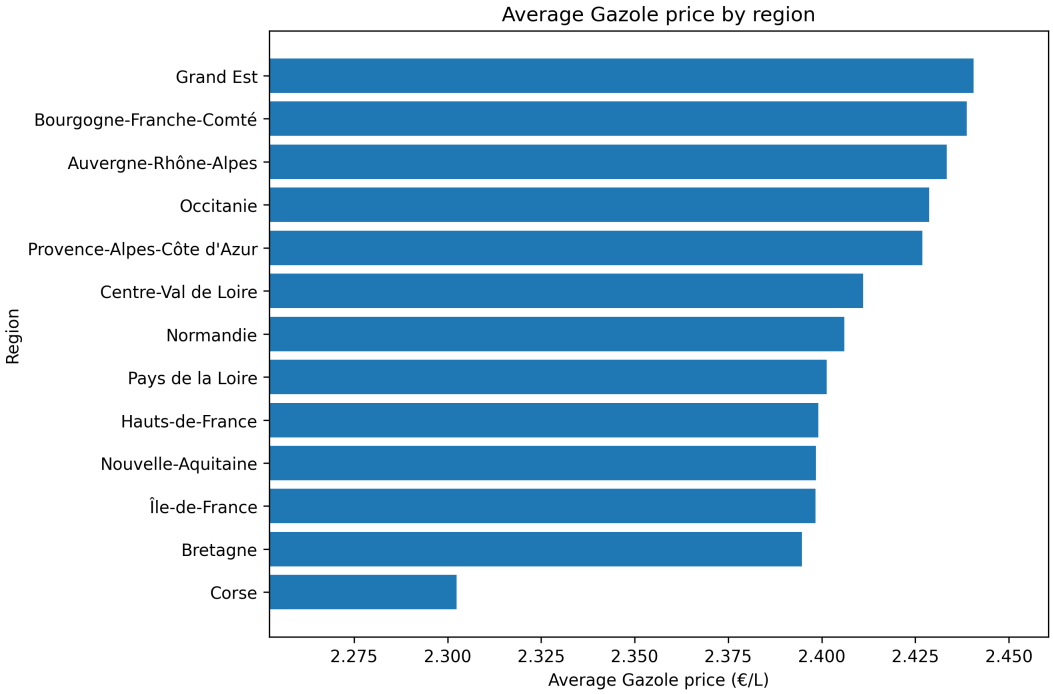


Figure 2: Average Gazole price by region

Average Gazole prices range from approximately **€2.302/L in Corse** to **€2.441/L in Grand Est**, corresponding to a difference of about **€0.139/L**. This within-fuel geographical variation motivates the inclusion of location controls in the econometric analysis.

Tentative Analysis Plan

We will estimate and compare several regression specifications. A general specification is

$$Price_i = \beta_0 + \beta' FuelType_i + \gamma' Location_i + \delta' StationCharacteristics_i + u_i$$

where fuel type and location are represented by sets of indicator variables.

The analysis will proceed progressively. We will first estimate a specification controlling for fuel type, then introduce geographical controls, and finally add observable station characteristics such as available services when appropriate. We will compare estimated coefficients and goodness of fit across specifications and use statistical inference tools, including individual and joint significance tests.

Main Limitation

The dataset is **observational**. Consequently, the estimated coefficients will primarily be interpreted as conditional **associations rather than causal effects**. Potential determinants such as local competition, local demand, wholesale supply conditions or other station characteristics may not be fully observed.